Weather Analytics Dashboard

Business Problem

Weather data is widely available through APIs; however, it is often delivered in raw JSON format, making it difficult for users to extract meaningful insights or identify trends. Analyzing weather conditions across multiple locations requires data transformation, visualization, and reporting capabilities that raw API responses do not provide. As a result, decision-making based on weather information can become inefficient and time-consuming.

Project Objective

The objective of this project is to develop a centralized analytics solution that transforms real-time weather data into actionable insights. By integrating weather data with Power BI, the project aims to provide a user-friendly platform for monitoring and analyzing key weather indicators.

Solution

This project utilizes the OpenWeather API to collect real-time weather data, which is then transformed and cleaned using Power Query. The processed data is loaded into Power BI, where interactive dashboards and visualizations are created to present weather information in an intuitive and meaningful way.

The dashboard includes:

- Real-time monitoring of temperature, humidity, and wind speed.
- Interactive KPI cards for quick performance tracking.
- Dynamic filters and slicers for location-based analysis.
- Comparative analysis of weather conditions across different cities.
- Visual representations of weather trends and patterns.

Business Impact

The Weather Analytics Dashboard converts raw API data into actionable business intelligence by providing:

- A centralized platform for weather data analysis.
- Improved accessibility and understanding of weather information.
- Faster identification of weather trends and conditions.
- Data-driven insights to support planning and decision-making.

- Enhanced reporting through interactive visualizations and dashboards.

Skills Demonstrated

- API Integration (OpenWeather API)
- Data Extraction and Transformation
- Data Cleaning using Power Query
- Data Modeling
- Power BI Dashboard Development
- KPI Design and Reporting
- Data Visualization
- Business Intelligence and Analytics

Tools & Technologies

- Power BI
- OpenWeather API
- Power Query
- Python
- REST API
- Data Visualization Techniques

Key Outcome

Successfully developed an interactive Power BI dashboard that transforms real-time weather data into meaningful insights, enabling efficient monitoring, comparison, and analysis of weather conditions across multiple locations.